ASSESSMENT-2

TOOL

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course code :- MCA0401

course :- Deep learning

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Introduction

Neural network is a machine learning model that learns pattern from data and predict output.

During training - model learns.

During testing - predicts result

Training MSE = 0.85

Testing MSE = 5.20

MSE (mean squared Error)

$$MSE = \frac{1}{n} \Sigma (Actual - predicted)^2$$

lower MSE - better predicted.

Higher MSE - large predicted Errors.

Given,

Training MSE = 0.85

Testing MSE = 5.20

$5.20 > 0.85$

the model has high testing error.

$\therefore$ It is overfitting.

Reasons for High Testing error

→ overfitting

→ Small training Dataset

→ Complex neural network.

→ noise in data.

→ lack of Regularization.

→ Improper data splitting.

corrective measures

→ Increase training data
→ use Regularization
→ Early stopping
→ Simplify the model
→ Remove noise
→ Tune Hyperparameters.

Training error	Testing error	Interpretation
High	High	underfitting
low	High	overfitting
low	low	Goodmodel

This model is overfitting.

conclusion :-

Since the testing error is much larger than the training error, the model is suffering from overfitting.